

ASHWIN RAVICHANDRAN

Fairfax, VA (Willing to Relocate)

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EDUCATION

George Mason University

Master's in Computer Science (GPA: 3.70)

Fairfax, VA

Aug 2024 – May 2026

SNS College of Technology

Bachelor's in Computer Science and Engineering (GPA: 3.3/4)

Coimbatore, India

Jun 2021 – May 2024

TECHNICAL SKILLS

Languages:	Python, JavaScript, TypeScript, SQL, Golang
AI/ML:	LLMs, RAG, NLP, Vector Search, TensorFlow Lite, Model Quantization
Backend:	FastAPI, Flask, REST APIs, API Design, JWT Authentication, Uvicorn
Cloud & DevOps:	AWS (EC2, S3, Lambda, API Gateway, IAM), Docker, CI/CD, Git/GitHub
Databases:	PostgreSQL, DynamoDB, MongoDB, Redis

PROJECTS

GMU Intelligent Resource Navigation AI Chatbot

- Engineered a RAG-based AI chatbot using FastAPI, Gemini API, and ChromaDB, achieving 95%+ intent recognition across 26 GMU campus resources with sub-second response times.
- Built a full-stack React frontend with a web scraping pipeline, sentence embeddings, and persistent analytics, deployed live via GitHub continuous deployment..
- Authored 58 automated unit, integration, and edge-case tests to improve reliability for ambiguous queries, misspellings, and production-style chatbot usage.

Personalized On-Device Human Activity Recognition on Android

- Trained a lightweight 1D-CNN on the WISDM dataset, achieving 96.92% accuracy, and integrated it as a live TFLite Android app on Samsung Galaxy S25 with sub-1 ms inference.
- Evaluated real-world phone placement domain shift, identifying a 75–82 percentage point accuracy drop, and recovered +17 pp through memory-efficient last-layer fine-tuning using only 5 minutes of personal calibration data.
- Applied INT8 post-training quantization, reducing model size $2.8\times$ (0.106 MB $\rightarrow$ 0.037 MB) with only 0.23 pp accuracy loss for efficient on-device deployment.

EXPERIENCE

Prompt Infotech

Backend & Cloud Developer Intern

Coimbatore, India

Jan 2024 – May 2024

- Developed and deployed backend services on AWS with scalable architecture and secure IAM configurations.
- Developed production-style REST APIs using FastAPI and DynamoDB for real-time data processing, validation, and client integrations.
- Optimized PostgreSQL query performance through schema design, indexing, and query tuning, reducing execution time by approximately 30% for frequently used queries..
- Containerized backend services with Docker in Linux environments and supported CI/CD workflows using GitHub Actions.

The Reciprocal Solution

Full-Stack Developer Intern

Remote

Dec 2021 – Jan 2022

- Delivered full-stack web features using React.js, Node.js, and MongoDB, from database schema design to responsive UI implementation.
- Built and integrated RESTful APIs for forms and authentication with clean JSON contracts and error handling.
- Supported cloud deployments and release workflows using AWS/Heroku and Git/GitHub.